

CHY1009

Chemistry and Environmental Studies

Module 2

Part 1

States of Matter, Thermodynamics, Kinetics and Catalysis

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States of Matter, Thermodynamics, Kinetics and Catalysis

- ☐ States of Matter
- ☐ Surface Chemistry
- ☐ Thermodynamics
- ☐ Chemical Kinetics
- ☐ Enzyme catalysis

States of Matter

• Solid Liquid Gas Plasma

- Based upon particle arrangement
- Based upon energy of particles
- Based upon distance between particles

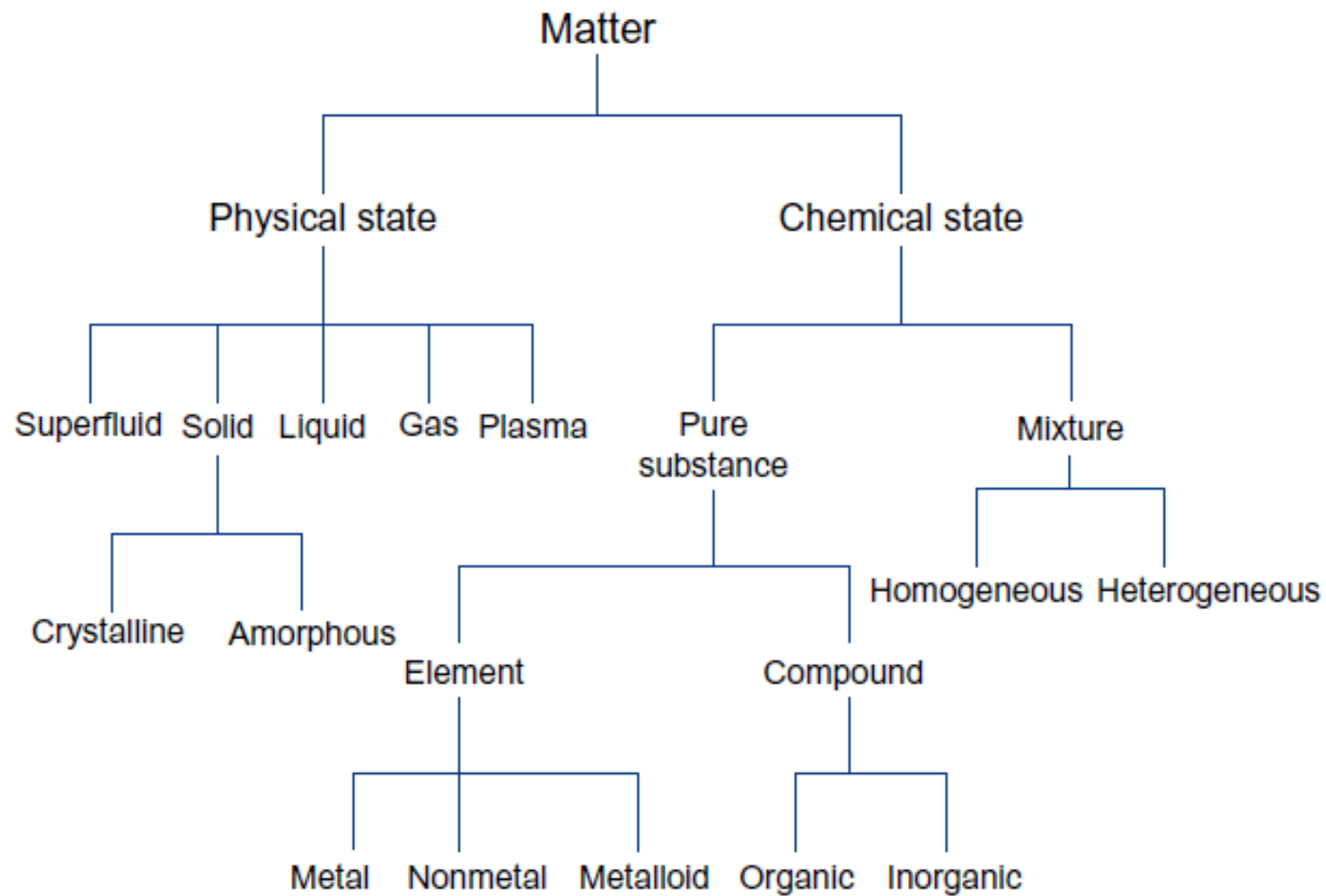
The Physical States of Matter -

Particulate theory of matter

- Greek philosopher Democritus described matter as made up of tiny **microscopic particles**.
- the particulate theory of matter was first presented by Democritus in 400 BC

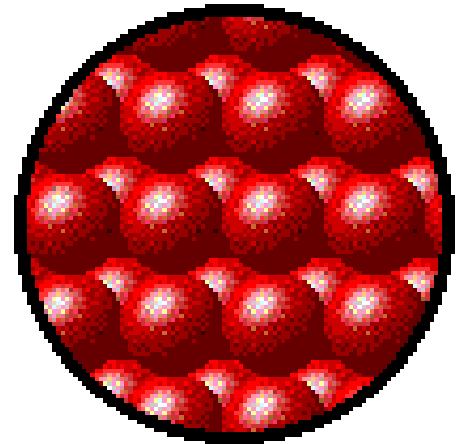
The Physical States of Matter - Particulate theory of matter

- **John Dalton** who is credited with first developing the modern **atomic theory of matter** in the early 1800s.
- The theory presented by **Democritus** was predominately a philosophical statement and therefore not widely accepted.
- In contrast, ***Dalton's theory*** was based on careful experimental observations and measurements and so became the first atomic theory of matter founded on scientific concepts and principles.
- According to the **modern atomic theory**, matter can be classified according to both its physical and chemical state



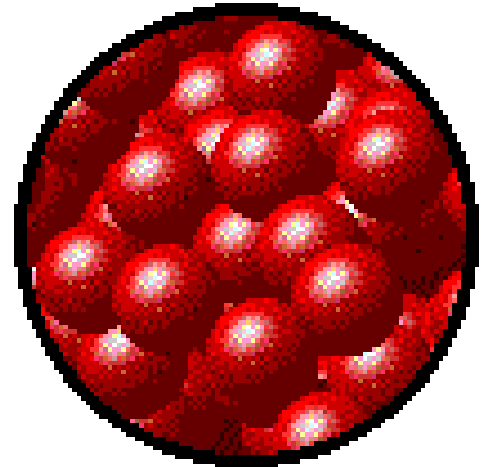
SOLIDS

- **Particles of solids are tightly packed, vibrating about a fixed position.**
- **Solids have a definite shape and a definite volume.**



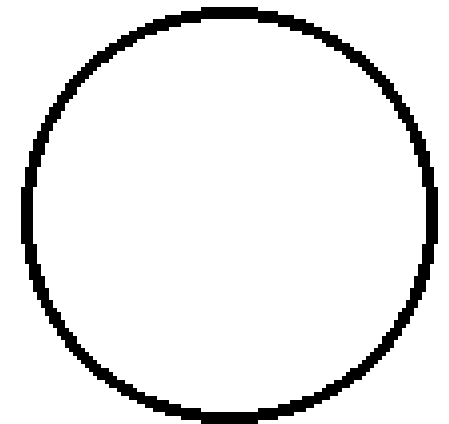
LIQUID

- Particles of liquids are tightly packed, but are far enough apart to slide over one another.
- Liquids have an indefinite shape and a definite volume.



GAS

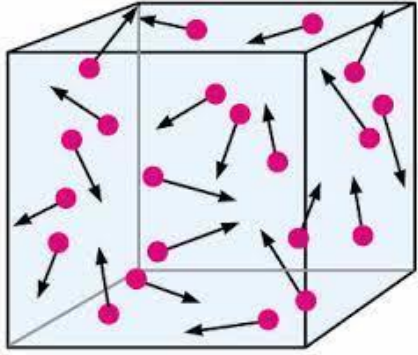
- **Particles of gases are very far apart and move freely.**
- **Gases have an indefinite shape and an indefinite volume.**



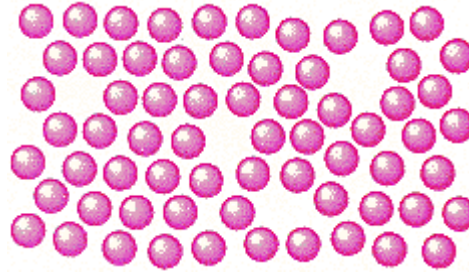
PHASE CHANGES

1. **Melting:** Solid to Liquid
2. **Freezing:** Liquid to Solid
3. **Evaporation:** Liquid to Gas
4. **Condensation:** Gas to Liquid
5. **Sublimation:** Solid to Gas
6. **Deposition:** Gas to Solid

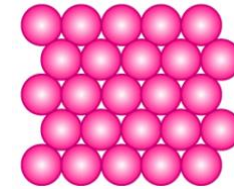
Physical States of Mater- Properties



Gas



Liquid



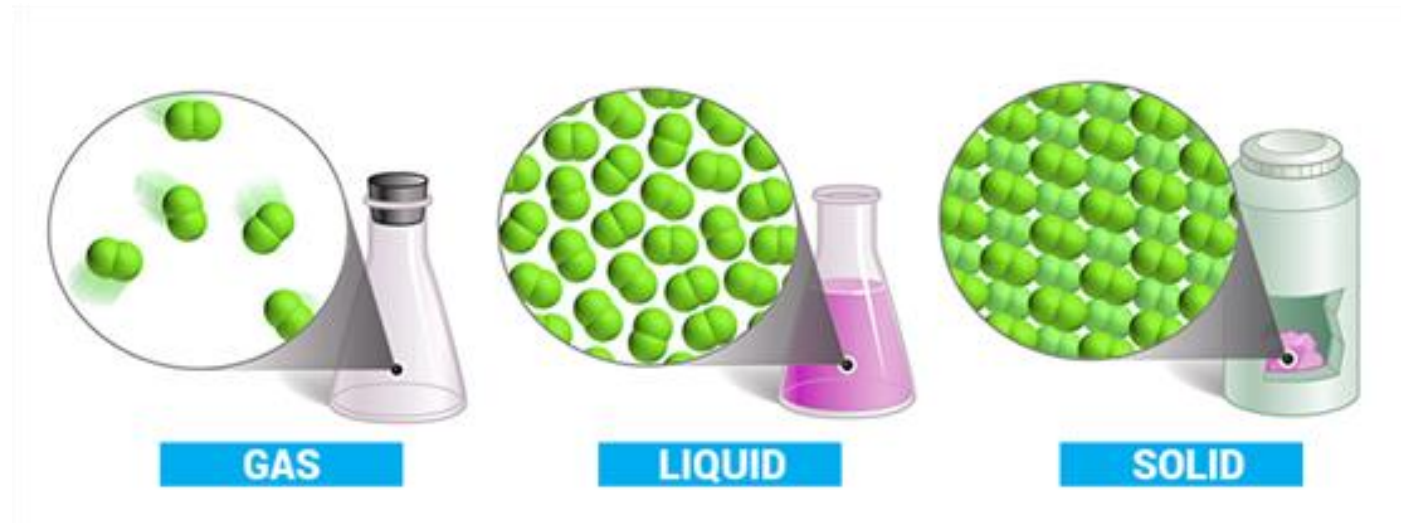
Solid

	Solid	Liquid	Gas
Packing	Tight	Close	Well separated
Free space	Little	Little	Large
Order	High	None	Random
Movement	Fixed	Slow	Rapid

Density (mass/cm³) of Gas, Liquid, Solids

$$\rho = \text{mass/unit volume} = M/V$$

- For a given mass of substance, density increases while going from **gas-to-liquid** and **liquid-to-solid**.



Interparticle distance decreases, this decreases volume. Density (ρ) $\propto$ 1/Volume (V)

Density (mass/cm³) of Gas, Liquid, Solids

	<i>Solid (g/cm³)</i>	<i>Liquid (g/cm³)</i>	<i>Gas (g/cm³)</i>
Ar	1.65	1.40	0.001784
N ₂	1.026	0.8081	0.001251
O ₂	1.426	1.149	0.001429

-the density of a typical solid is about 20% larger than the corresponding liquid,

-while the liquid is roughly 800 times as dense as the gas.



But what happens if you raise the temperature to super-high levels... between 1000°C and $1,000,000,000^{\circ}\text{C}$?

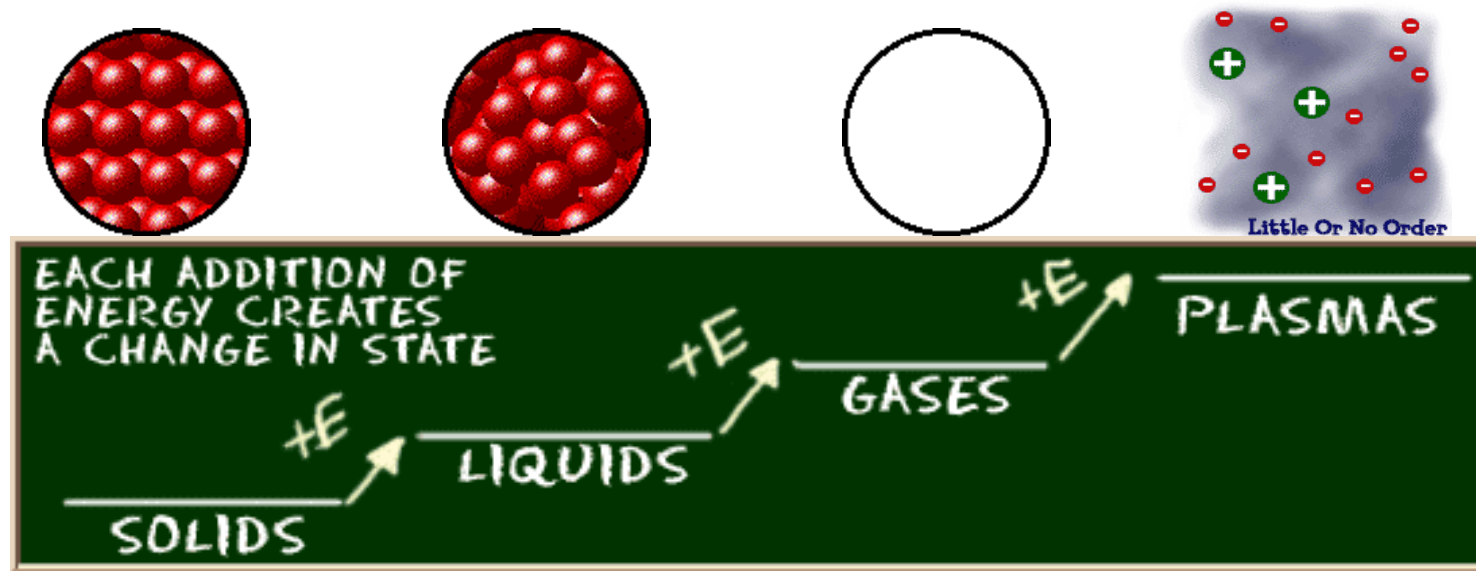
PLASMA

- A plasma is an ionized gas.
- A plasma is a very good conductor of electricity and is affected by magnetic fields.
- Plasmas, like gases have an indefinite shape and an indefinite volume.



PLASMA

- ❑ Plasma is considered **4th State of Matter** apart from solids, liquids and gases. It is one of the fundamental states of matter.
- ❑ Technically, it is an **ionized gas** consisting of positive ions and free electrons ,typically at low pressures (as in the upper atmosphere and in fluorescent lamps) or at very high temperatures (as in stars and nuclear fusion reactors).
- ❑ A plasma is a very **good conductor** of electricity. Example: Fluorescent lights, stars
- ❑ Like gases, plasma have an **indefinite shape and an indefinite volume**.
- ❑ It is most common (99.9 %) state of matter in the **universe**.



Tightly packed,
in a regular
pattern
Vibrate, but do
not move from
place to place

Close together
with no regular
arrangement.
Vibrate, move
about, and slide
past each other

Well separated
with no regular
arrangement.
Vibrate and
move freely at
high speeds

Has no definite
volume or shape
and is composed
of electrical
charged particles

Some places where plasmas are found...



Flame



Some places where plasmas are found...



Aurora (Northern Lights)

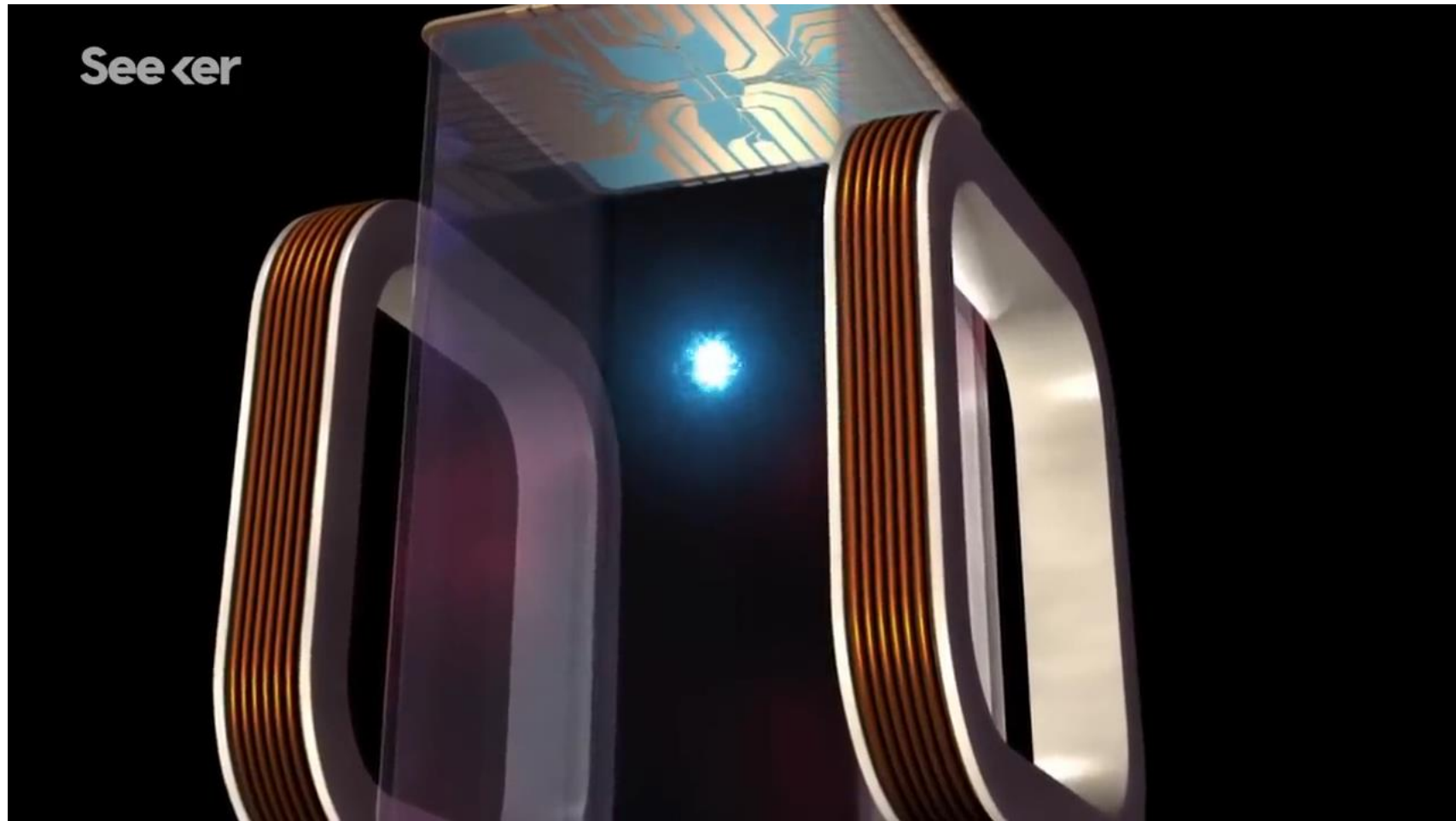


Sun: star in its plasma state

Physical changes

- **Physical changes** occur when objects undergo a change that does not change their chemical nature.
- A physical change involves a change in physical properties.
- Physical properties can be observed without changing the type of matter.
- Examples of physical properties include: texture, shape, size, color, odor, volume, mass, weight, and density.

5TH State?



<https://www.youtube.com/watch?v=VGKNWTOPsr4>